ABES Engineering College, Ghaziabad

Department of Applied Sciences & Humanities

Session: 2025-26 **Semester: I**

Course Code: 25AS102

Course Name: Fundamentals of Linear Algebra & Statistics

Assignment 2

S.No.	KL	CO	Question
1	K3	CO1	Show that $Q(\sqrt{2}) = \{a + b\sqrt{2} : a, b \epsilon Q\}$ is a vector space over Q with respect to the compositions $(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c + (b + d)\sqrt{2})$ $\text{k}a + b\sqrt{2}) = ka + kb\sqrt{2}, \quad where\ a, b, c, d\ \epsilon\ Q.$
2.	K3	CO1	Show that every plane passing through origin is a subspace of R^3over R $that\ is$ $W = \{(a, b, c): la + mb + nc = 0; l, m, n \in R, a, b, c \in R\}$ is a subspace of R^3over R.
3.	K3	CO1	Show that the set S = {(1,0,0), (0,1,0), (0,0,1), (1,1,1)}. Is not a basis for $R^3(R)$.
4	K3	CO1	Let V be the vector space of square matrices over R and let W be the sub space generated by $\begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix}, \begin{bmatrix} 2 & -4 \\ -5 & 7 \end{bmatrix}, \begin{bmatrix} 1 & -7 \\ -5 & 1 \end{bmatrix}$ Show that $\{\begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}\}$ form a basis and dim W=2.
5	K3	CO1	In the vector space of polynomial of degree<= 4, which of the following sets are linearly independent? (i) $x + 1, x^3 - x + 1, x^3 + 2x + 1$ (ii) $x^3 + 1, x^3 - 1, x^4 - x,\ x$ (iii) $x + 1,\ x + x^2, x^2 + x^3, x^3 + x^4, x^4 - 1$

Answers:

5.(i) linearly independent

(ii) linearly independent

(iii) linearly dependent